Higher Colleges of Technology

كليات التقنية العليا

CDS 2413- PROGRAMMING FOR DATA SCIENCE

Week 9-10

Week 10	CLO 3: Create scripts to solve business problems using data analysis techniques such as classification, clustering, optimization, and recommendation

Document Revision CONTROL (DRC)

Version	Author	Effective Date	Change Description	DRC No
1.0	Mohammed Abdul Rahim	2/05/2023	Define the first version	001

X.Y
X: Major change
Y: minor change

Lecture Outline

- Classification and Model Selection
- Logistic Regression
- K-Nearest Neighbours-
- Naïve Bayes
- Decision Tree
- Python Programming for Classification

LECTURE OBJECTIVES

- The student should be able to do the following at the end of the lecture:
  - Understand the **basic concepts** of classification models.
  - Implement **classification models** using Python.
  - Interpret the **results** of classification models.
  - Apply **classification models** to real-world problems.

3. Classification – Naïve Bayes

Classification – Bayes Theorem

- Bayes' theorem is a fundamental concept in probability theory that describes how to update our beliefs about an event (hypothesis) when we receive new information (evidence).

نظرية بايز هي مفهوم أساسي في نظرية الاحتمالات يصف كيفية تحديث معتقداتنا حول حدث ما (الفرضية) عندما نتلقى معلومات جديدة (الأدلة).

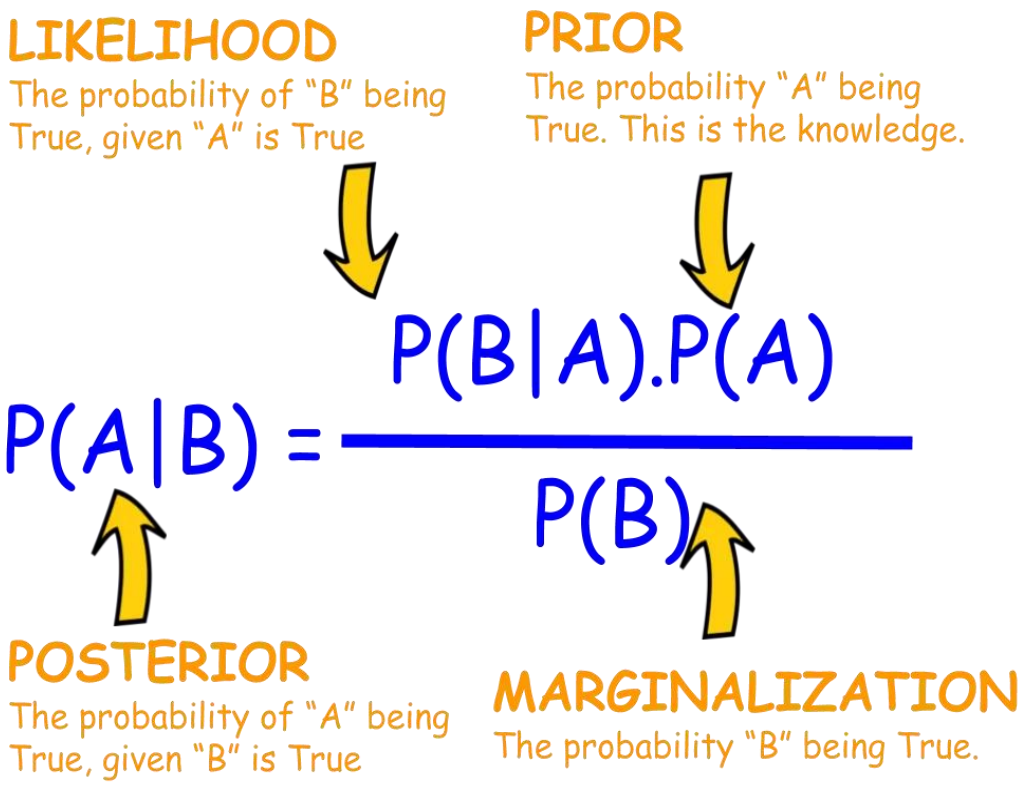
Classification – Bayes Theorem

- **Why Naïve Bayes?**
  - Simple to implement
  - Works well with high-dimensional data
  - Requires a small amount of training data
  - Handles both continuous and discrete data
  - Effective in various applications such as spam filtering, text classification, and medical diagnosis

Classification – Bayes Theorem

- To implement Bayes' theorem, we can use the following steps:
  - Define the prior probability of the event (hypothesis) P(A).
  - Define the likelihood of the evidence given the event P(B|A).
  - Calculate the probability of the evidence P(B).
  - Use Bayes' theorem formula to calculate the posterior probability P(A|B).

$$P(A|B) = (P(B|A) * P(A)) / P(B)$$



Classification – Bayes Theorem - Example

- Machine M1 produces 30 toys per hour.
- Machine M2 produces 20 toys per hour.
- Among all produced toys, 1% are defective (D).
- Of all defective toys:
  - 50% are from M1.
  - 50% are from M2.
- What is the probability that a toy produced by M2 is defective (D)?

Toy Machine 1 (M1)

Toys produced by M1

Toy Machine 2 (M2)

Toys produced by M2

Classification – Bayes Theorem

- M1 : 30 toys / hr
- M2: 20 toys / hr
- Out of all produced toys:
  - Defective(D)=1%
- Out of all defective toys:
  - 50% from M1
  - 50% from M2

- What is the probability that a toy produced by M2 is defective(D)?

- $P(M1) = 30 / 50 = 0.6$
- $P(M2) = 20 / 50 = 0.4$
- $P(D) = 1\% = 0.01$
- $P(M1|D) = 50\% = 0.5$

  $P(M2|D) = 50\% = 0.5$

- $P(D|M2) = ?$

Classification – Bayes Theorem

- P(M1) = 30 / 50 = 0.6
- P(M2) = 20 / 50 = 0.4
- P(D) = 1% = 0.01
- P(M1|D) = 50% = 0.5

  P(M2|D) = 50% = 0.5

- P(D|M2)= ?

- According to Bayes theorem

$$P(D|M2) = \frac{P(M2|D) * P(D)}{P(M2)}$$

$$= \frac{0.5 * 0.01}{0.4}$$

$$= 0.0125$$

- For 1000 toys, 12.5 toys are defective from M2

Classification – Bayes Theorem

- What is the probability that a toy produced by M1 is defective(D)?

P(D|M1)= ?

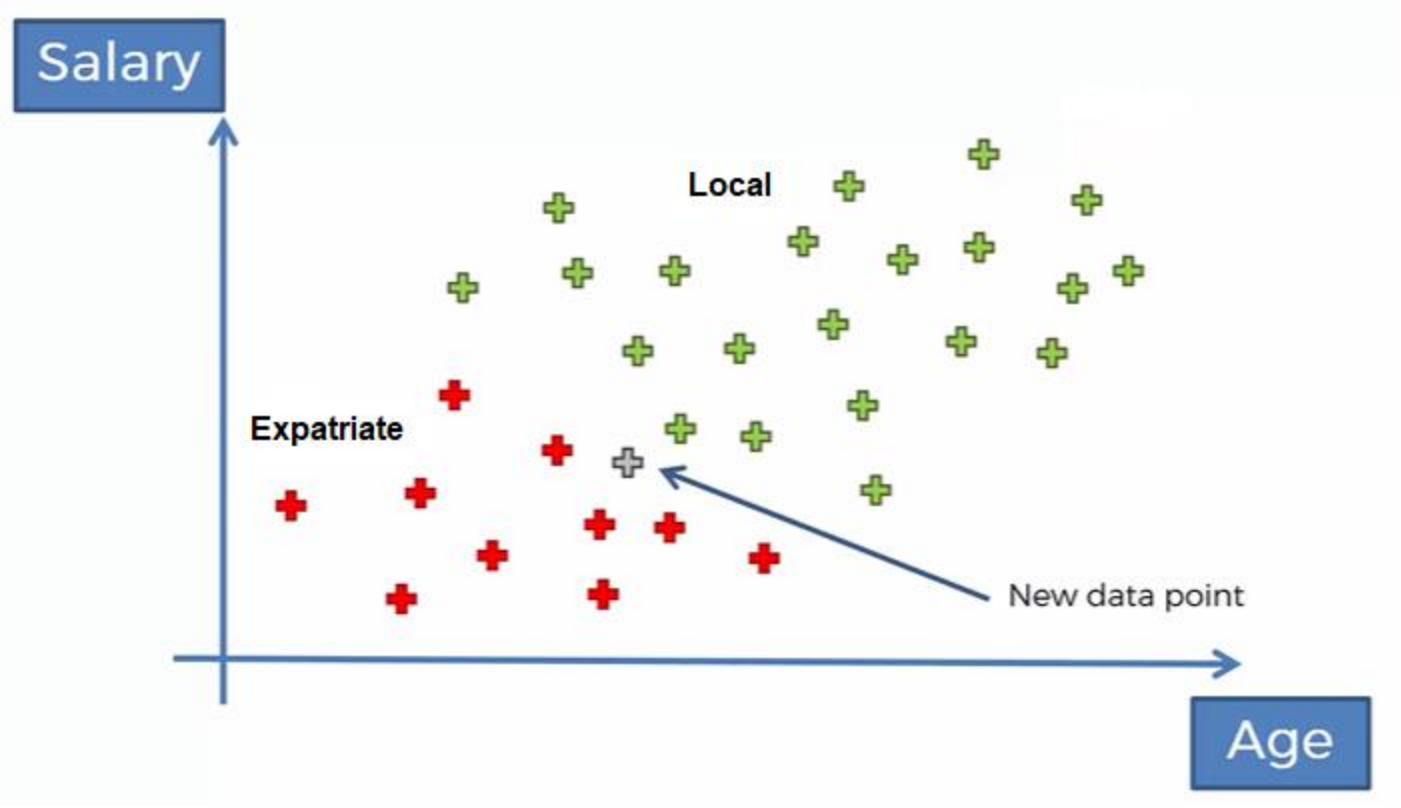
- P(M1) = 30 / 50 = 0.6
- P(M2) = 20 / 50 = 0.4
- P(D) = 1% = 0.01
- P(M1|D) = 50% = 0.5

  P(M2|D) = 50% = 0.5

- P(D|M1)= ?

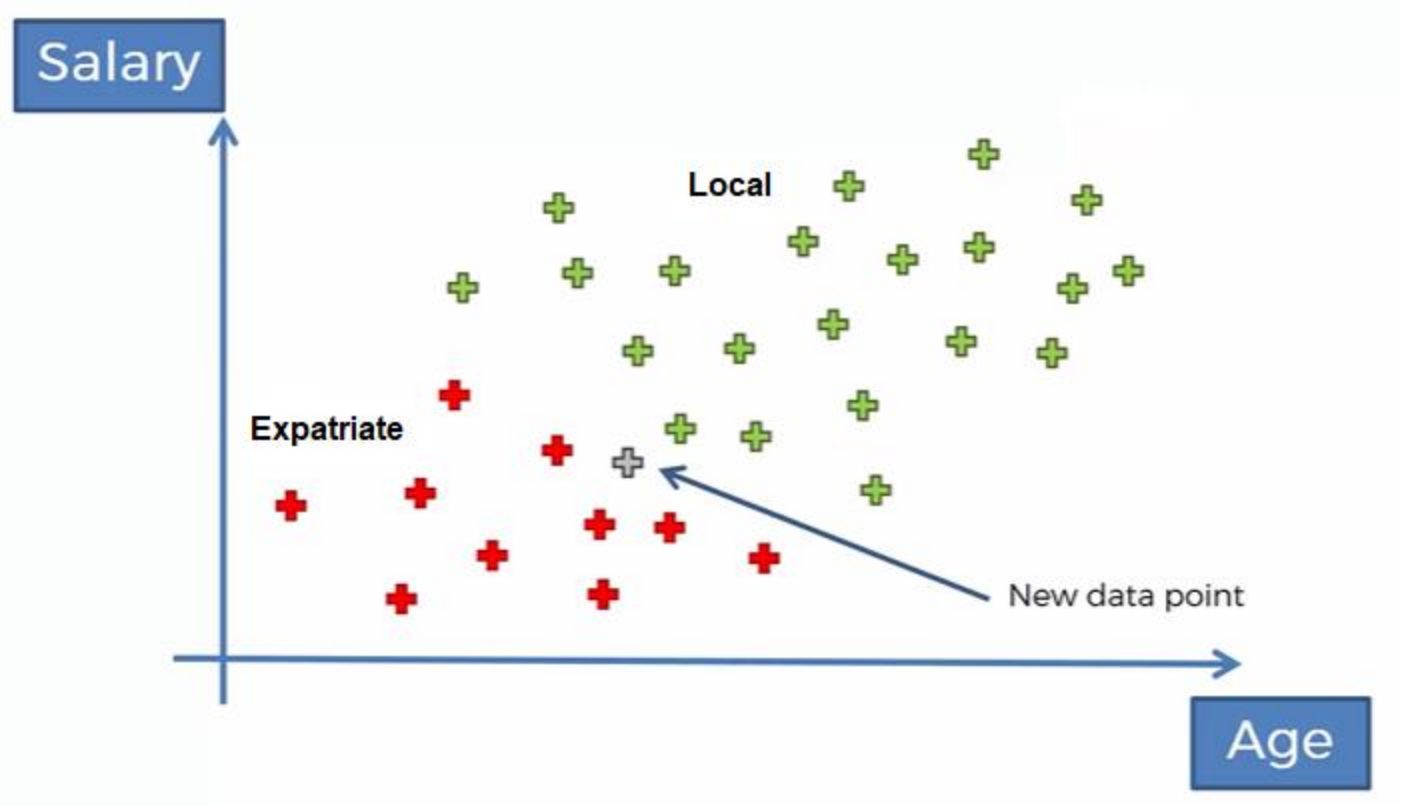
Classification – Naïve Bayes Theorem

- Consider the scatter plot between age and salary is plotted
- There are two categories:
  - Local and
  - Expatriate
- If new data point arrives, then which category new data point will be placed.



Classification – Naïve Bayes Theorem

- Apply Naïve Bayes theorem to find where new data point has to be placed in Local (L) or Expatriate (E)
- Find P(E|X) and P(L|X)
- X is the feature which can be age or salary



Classification – Naïve Bayes Theorem

- According to Naïve Bayes theorem
- $P(E|X) = \frac{P(X|E) * P(E)}{P(X)}$
- $P(L|X) = \frac{P(X|L) * P(L)}{P(X)}$

Posterior Probability

Likelihood

Prior Probability

$$P(E|X) = \frac{P(X|E) * P(E)}{P(X)}$$

Posterior Probability

Likelihood

Prior Probability

Marginal Likelihood

$$P(L|X) = \frac{P(X|L) * P(L)}{P(X)}$$

Marginal Likelihood

Calculate posterior probabilities to determine the class of the new data point.

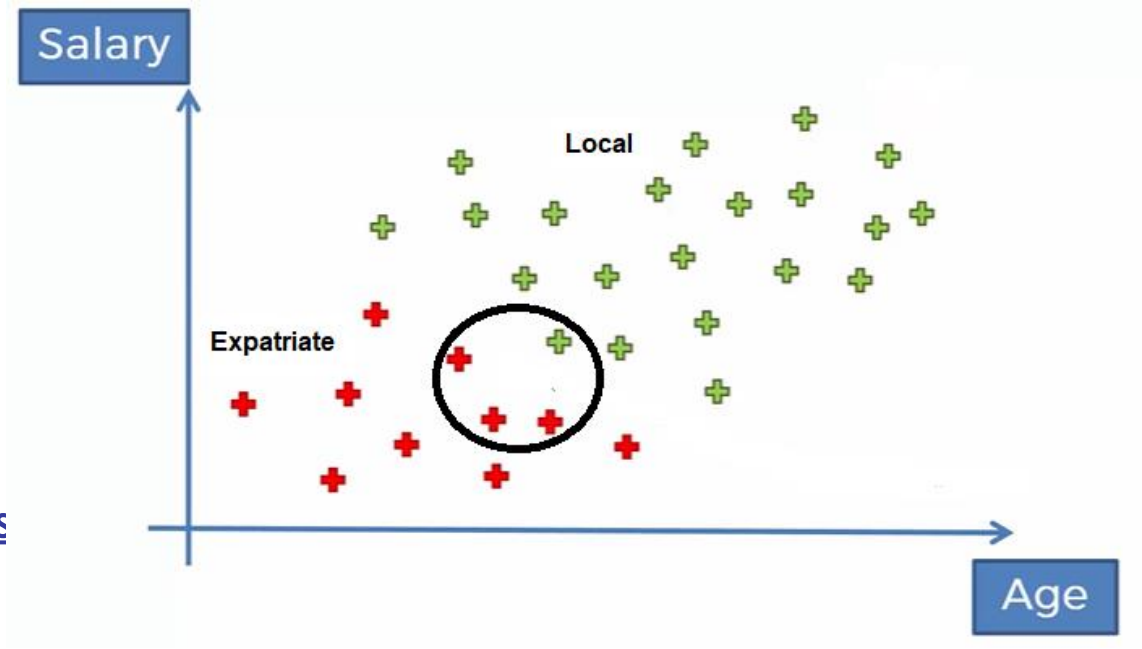
Classification – Naïve Bayes Theorem

- $P(E) = \frac{\text{Number of Expatriates}}{\text{Total observation}}$

  $= 10 / 30$
- $P(X) = \frac{\text{Number of Similar Observation}}{\text{Total observation}}$

  $= 4 / 30$
- $P(X|E) = \frac{\text{Number of Similar Observation for Expatriates}}{\text{Total number of Expatriates}}$

  $= 3 / 10$



Classification – Naïve Bayes Theorem

- According to Naïve Bayes theorem
- $P(E|X) = \frac{P(X|E) * P(E)}{P(X)} = \frac{3/10 * 10/30}{4/30} = 0.75$
- $P(L|X) = \frac{P(X|L) * P(L)}{P(X)} = \frac{1/20 * 20/30}{4/30} = 0.25$
- Thus, 75% of expatriate and 25% of Locals exhibits X features

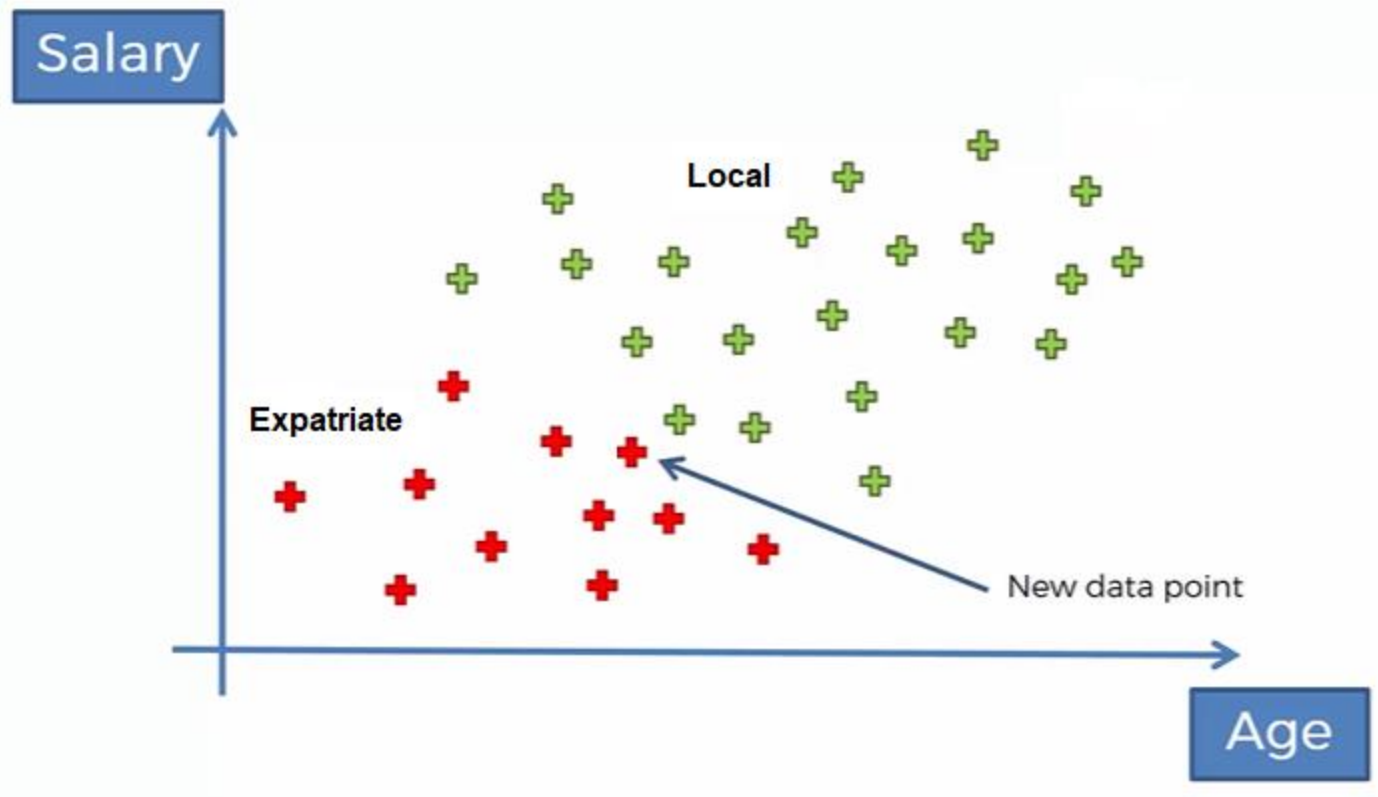
- $P(E) = 10 / 30$
- $P(X) = 4 / 30$
- $P(X|E) = 3 / 10$
- $P(L) = 20 / 30$
- $P(X|L) = 1 / 20$

Classification – Naïve Bayes Theorem

- Thus,

  $P(E|X) > P(L|X)$

- So, new data point should be added to Expatriate category



Build, Train, Predict and Evaluate the Naïve Bayes Model– Case Study

Classification – Naïve Bayes - Python

Naive Bayes

- Importing the libraries
- Importing the dataset
- Splitting the dataset into the Training set and Test set
- Feature Scaling
- Training the Naive Bayes model on the Training set
- Making the Confusion Matrix

Similar for all the models

Classification – Naïve Bayes - Python

Training the Naive Bayes model on the Training set

```
[ ] from sklearn.naive_bayes import GaussianNB
    classifier = GaussianNB()
    classifier.fit(X_train, y_train)
```

Attaching the classifier to the train dataset

Configuring for Naïve Bayes model by mentioning the classifier is Naïve Bayes

Classification – Naïve Bayes - Python

Making the Confusion Matrix

```
from sklearn.metrics import confusion_matrix, accuracy_score
y_pred = classifier.predict(X_test)
cm = confusion_matrix(y_test, y_pred)
print(cm)
accuracy_score(y_test, y_pred)
```

```
[[99  8]
 [ 2 62]]
0.9415204678362573
```

False Positive(FP)

False Negative(FN)

Accuracy Rate (94%)

Number of Errors = FP + FN = 8+ 2 = 10

Classification – Naïve Bayes - Python

Predicting the value

```
[38] print(classifier.predict(sc.transform([[1173347,1,1,1,1,2,5,1,1,1]])))
```

```
[2]
```

It predicted correctly that class=2. So, the model is working fine.

```
print(X_test)
```

```
[[1173347       1       1 ...       1       1       1]
 [1156017       3       1 ...       2       1       1]
 [ 706426       5       5 ...       4       3       1]
```

```
print(y_test)
```

```
[2 2 4 4 2 2 2 4 2 2 4 2 4 2 2 2 4 4 4 2 2 2 4 2 4 4 2 2 2 4 2 4 4 2 2 2 4
 4 2 4 2 2 2 2 2 2 2 4 2 2 4 2 4 2 2 2 4 2 2 4 2 2 2 2 2 2 2 2 4 4 2 2 2 2
 2 2 4 2 2 2 4 2 4 2 2 4 2 2 4 2 4 2 4 4 4 2 4 4 4 2 2 2 4 4 2 2 4 4 2 2 4
 2 2 4 2 2 2 4 2 2 2 4 2 2 4 4 2 4 2 4 2 2 4 2 2 4 2 4 2 2 2 4 2 2 2 2 4 2
 4 2 4 4 2 2 2 2 4 4 2 4 4 4 4 4 4 2 2 2 2 2 2]
```

Check in BreastCancer.csv file if data is not fully visible

4. Classification – Decision Tree

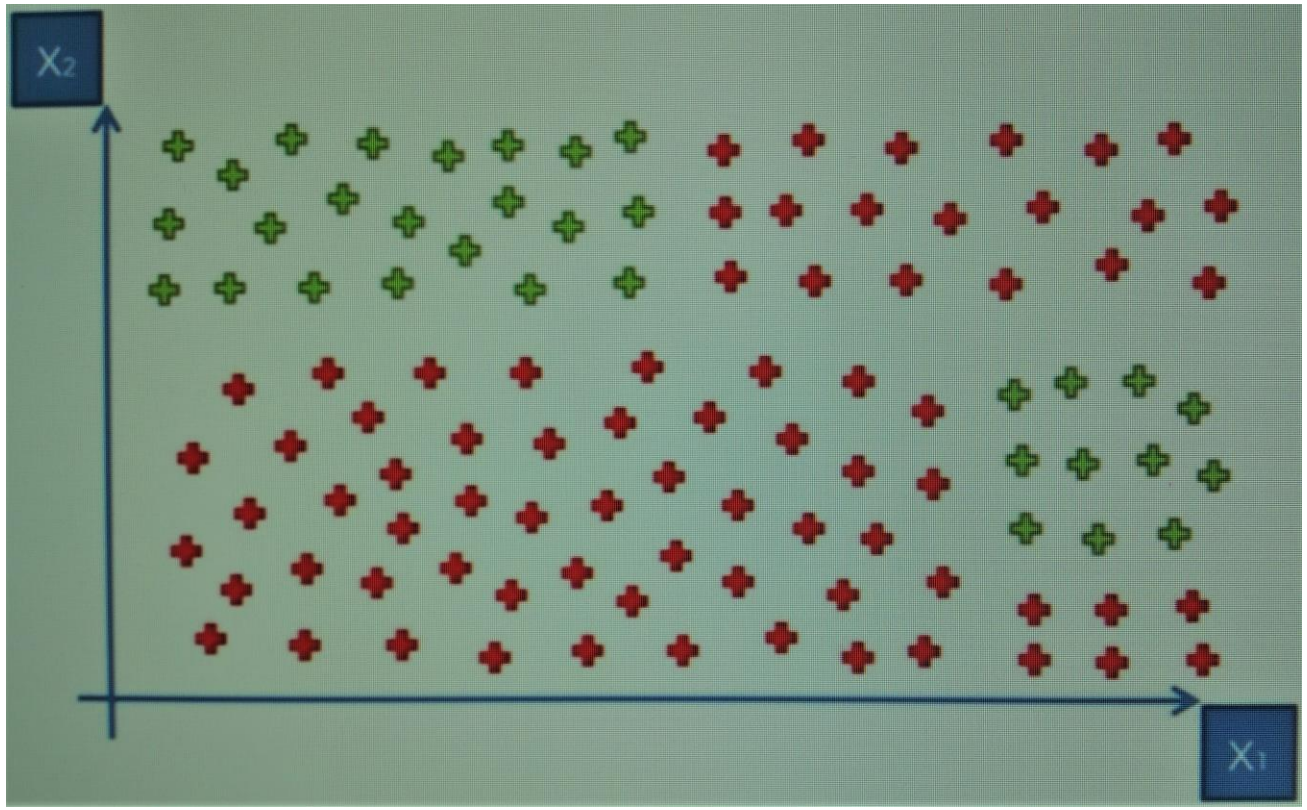
Classification – Decision Tree

- Decision trees are **tree-like structures** that represent a series of **decisions** and their possible **consequences**.
- They are **easy to understand and interpret**, making them a popular choice for **machine learning** applications.
- Decision trees offer a visual and intuitive way to make decisions based on data. Understanding both their advantages and limitations is crucial for effectively applying them in machine learning.

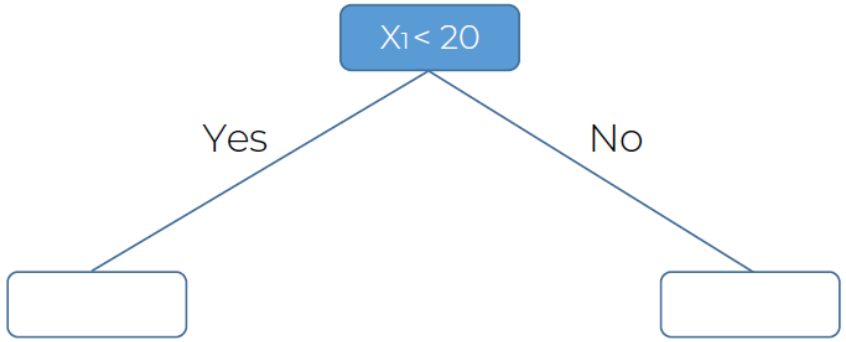
- **Advantages:**
- Can handle both **categorical** and **numerical data**.
- **Robust** to outliers.
- **Disadvantages:**
- Can be prone to **overfitting**.
- Sensitive to small changes in the data.
- May not perform well on **complex datasets**.

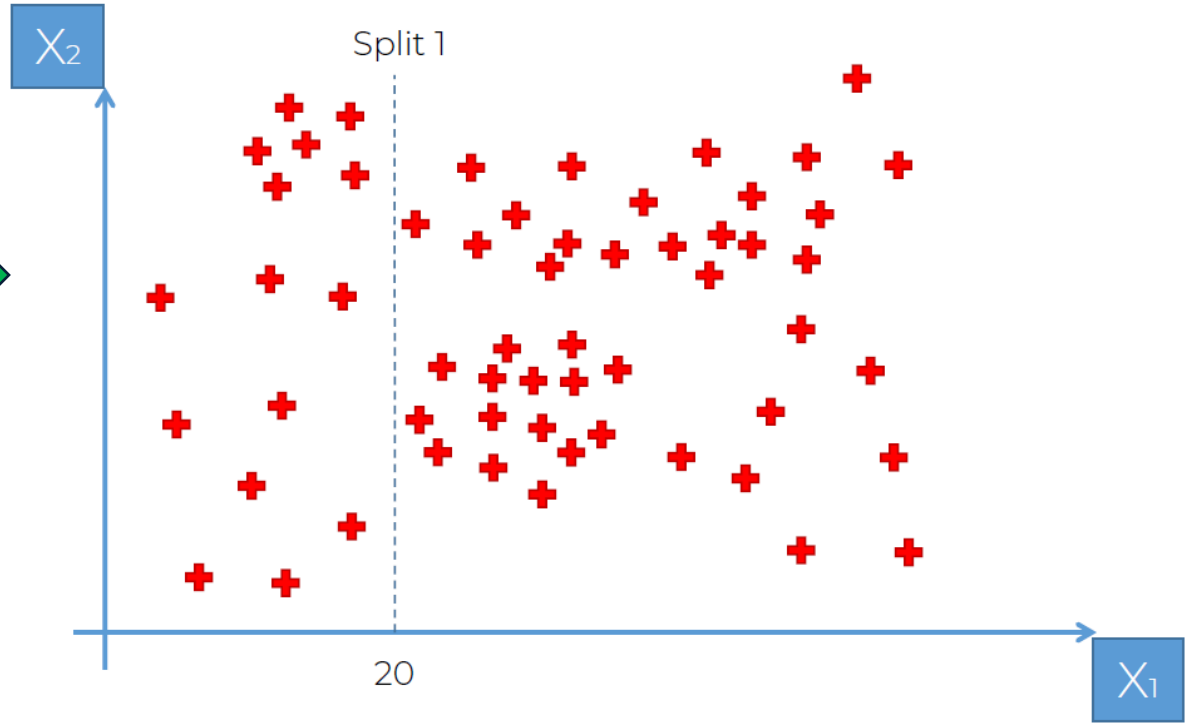
Classification – Decision Tree

- Consider the following dataset with scatter plot. How will you classify the data points based on the decision tree?



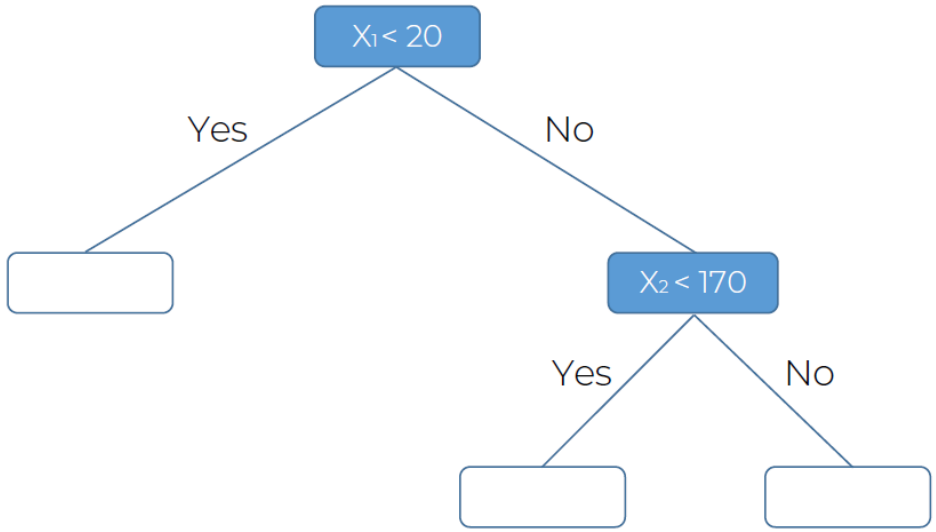
Classification – Decision Tree

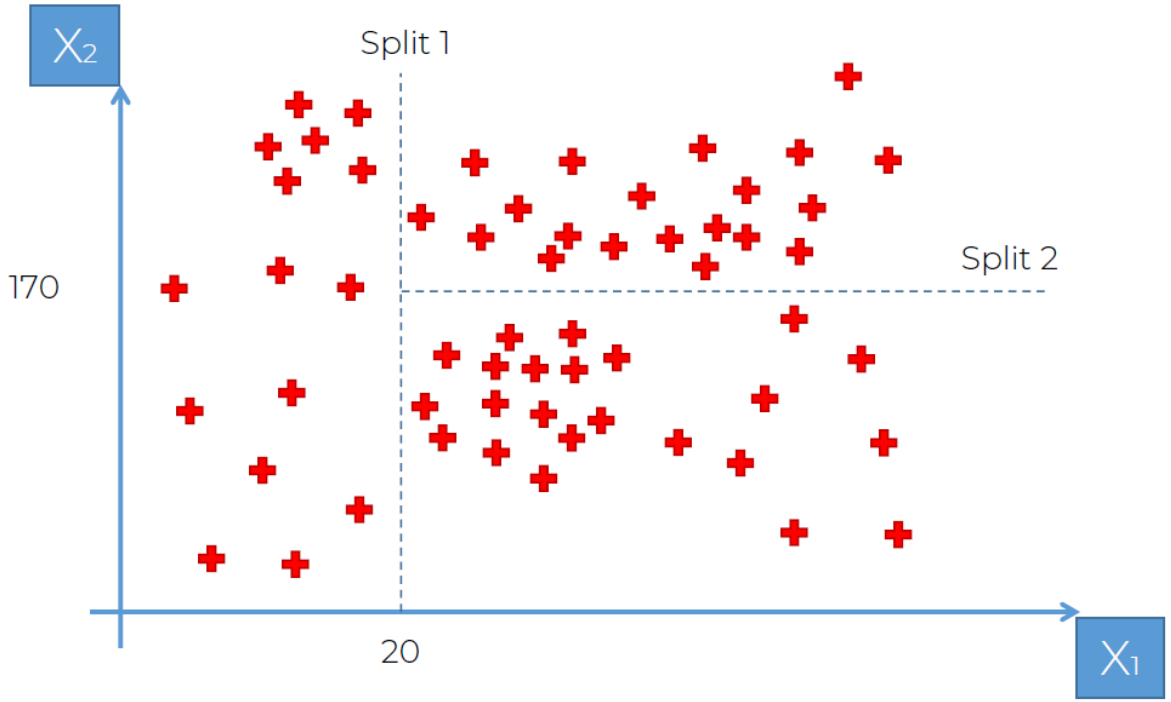




www.superdatascience.com

Classification – Decision Tree





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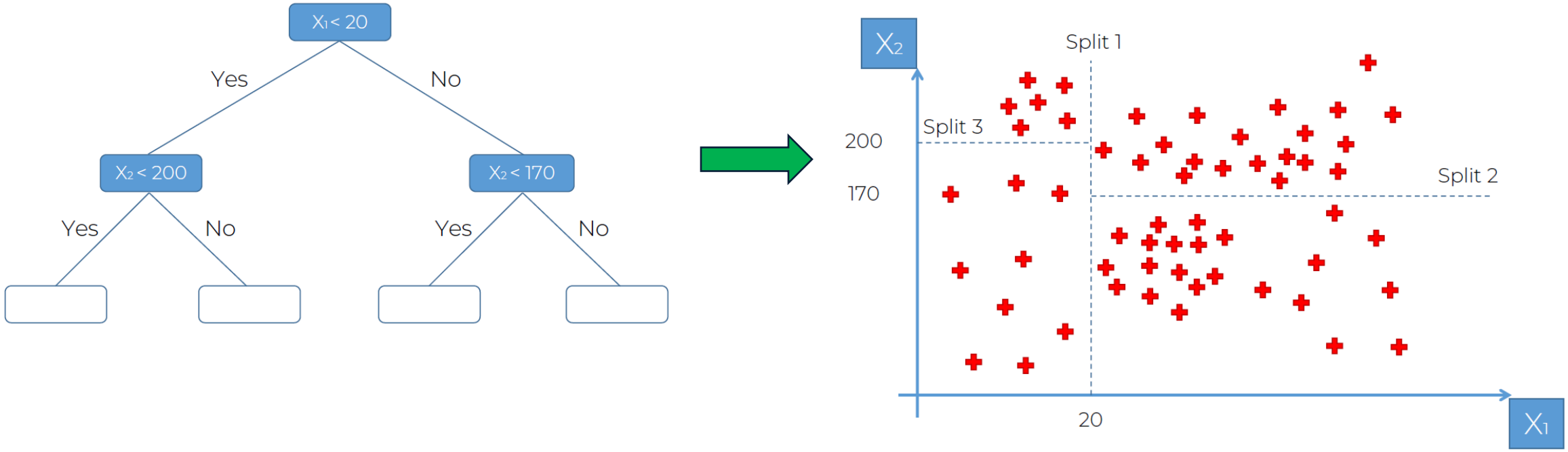
Classification – Decision Tree



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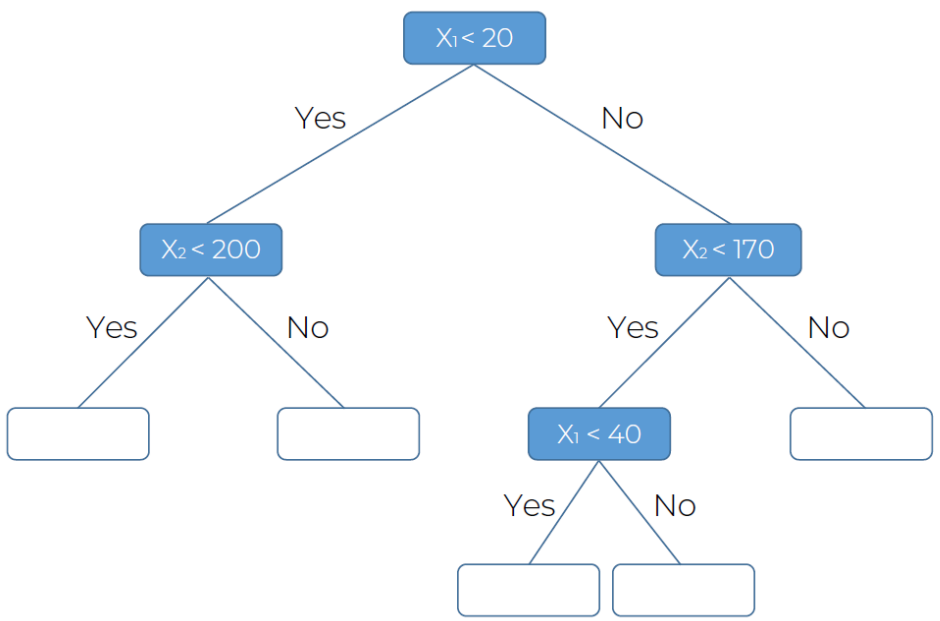
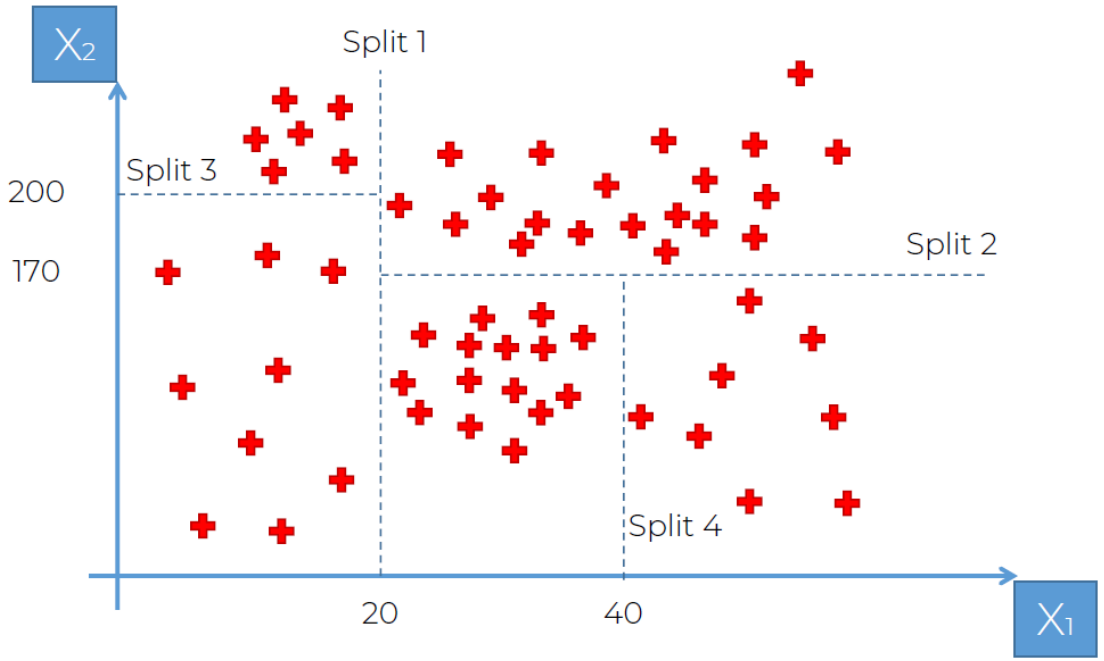
Classification – Decision Tree





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Build, Train, Predict and Evaluate the Decision Tree Model– Case Study

Classification – Decision Tree - Python

Decision Tree Classification

Importing the libraries

Importing the dataset

Splitting the dataset into the Training set and Test set

Feature Scaling

Similar for all the models

Training the Decision Tree Classification model on the Training set

Making the Confusion Matrix

Classification – Decision Tree - Python

Training the Decision Tree Classification model on the Training set

```
[ ] from sklearn.tree import DecisionTreeClassifier
    classifier = DecisionTreeClassifier(criterion = 'entropy', random_state = 0)
    classifier.fit(X_train, y_train)
```

Attaching the classifier to the train dataset

Configuring for Decision Tree model by mentioning the classifier is Decision Tree

Classification – Decision Tree - Python

Making the Confusion Matrix

```
from sklearn.metrics import confusion_matrix, accuracy_score
y_pred = classifier.predict(X_test)
cm = confusion_matrix(y_test, y_pred)
print(cm)
accuracy_score(y_test, y_pred)
```

```
[[103   4]
 [  3  61]]
0.9590643274853801
```

False Positive(FP)

False Negative(FN)

Accuracy Rate (94%)

Number of Errors = FP + FN = 4+ 3 = 7

Classification – Decision Tree - Python

Predicting the value

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[38] print(classifier.predict(sc.transform([[1173347,1,1,1,1,2,5,1,1,1]])))
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print(y_test)
```

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 2 2 4 2 2 2 4 2 4 2 2 4 2 2 4 2 4 2 4 4 4 2 4 4 4 2 2 2 4 4 2 2 4 4 2 2 4
 2 2 4 2 2 2 4 2 2 2 4 2 2 4 4 2 4 2 4 2 2 4 2 2 4 2 4 2 2 2 4 2 2 2 2 4 2
 4 2 4 4 2 2 2 2 4 4 2 4 4 4 4 4 4 2 2 2 2 2 2]
```

Check in BreastCancer.csv file if data is not fully visible

Classification – Evaluation

- Below is the result of performance metrics(confusion matrix) for classification models of the given data set:

Model	Errors	Accuracy
LR	9	94.73%
KNN	9	94.73%
NB	10	94.15%
DT	7	95.90%

Evaluate which is the best model for the given dataset?

Classification – Evaluation

- Below is the result of performance metrics(confusion matrix) for classification models of the given data set:

Model	Errors	Accuracy
LR	9	94.73%
KNN	9	94.73%
NB	10	94.15%
DT	7	95.90%

Thus, Decision Tree model is best fit model for the given dataset.

Project

15.	Predict the value of the dependent variable from the different classifier such as Logistic Regression, KNN, Naïve-Bayes and Decision Tree.
16.	Evaluate the performance of each model using confusion matrix and accuracy and identify the best fit classifier for the chosen dataset.
17.	Predict the dependent variable by using best-fit classifier.

Skill Yourself – Other Classification Models

Classification Model	Description	Evaluation Methods
Logistic Regression	Predicts the probability of a binary outcome (e.g., spam/not spam, yes/no).	Accuracy, Precision, Recall, F1-score, AUC-ROC, Log Loss
K-Nearest Neighbors (KNN)	Classifies data points based on the majority class among its k nearest neighbors.	Accuracy, Precision, Recall, F1-score
Naive Bayes	Based on Bayes' theorem with the assumption of independence between features.	Accuracy, Precision, Recall, F1-score, AUC-ROC
Decision Tree	Creates a tree-like model of decisions and their possible consequences to classify data.	Accuracy, Precision, Recall, F1-score, AUC-ROC
Support Vector Machines (SVM)	Finds an optimal hyperplane to separate data points into different classes.	Accuracy, Precision, Recall, F1-score, AUC-ROC
Random Forest	An ensemble learning method that combines multiple decision trees to improve accuracy.	Accuracy, Precision, Recall, F1-score, AUC-ROC
Neural Networks	Simulates the human brain's neural network to learn complex patterns.	Accuracy, Precision, Recall, F1-score, AUC-ROC
Convolutional Neural Network (CNN)	A type of deep learning model specifically designed for image recognition.	Accuracy, Precision, Recall, F1-score, AUC-ROC
Recurrent Neural Network (RNN)	A type of neural network designed to process sequential data, such as text or time series.	Accuracy, Precision, Recall, F1-score, AUC-ROC
Multilayer Perceptron (MLP)	A feedforward artificial neural network with multiple layers of perceptrons.	Accuracy, Precision, Recall, F1-score, AUC-ROC
AdaBoost	An adaptive boosting algorithm that combines multiple weak learners into a strong learner.	Accuracy, Precision, Recall, F1-score, AUC-ROC
Gradient Boosting	An ensemble learning method that builds an additive model in a forward stage-wise fashion.	Accuracy, Precision, Recall, F1-score, AUC-ROC

Do It Yourself - I

- Develop and evaluate classification models(Marks.CSV) :
  - To predict student pass/fail status based on their performance in 'Quiz', 'Practical', and 'Project' exams.
  - A student is considered 'Pass' if their **'Total' marks exceed 60**.
  - Implement and compare classification models such as Logistic Regression, K-Nearest Neighbors (KNN), Support Vector Machines (SVM), Decision Trees, and Random Forest.
  - Evaluate model performance using metrics like accuracy, precision, recall, and F1-score. Select the best-performing model based on these metrics.
  - Print the evaluation metrics for each model and determine the number of students predicted to pass and fail by each model.
  - *If the initial model performance is unsatisfactory, consider adjusting the dataset to improve classification accuracy.*

Do It Yourself - I

- Expected Output

```
Number of Students Passed: X
Number of Students Failed: Y

Evaluation Metrics for All Models:
Model                  Accuracy    Precision    Recall    F1-Score
Logistic Regression     0.85       0.83         0.88      0.85
Decision Tree           0.80       0.78         0.83      0.80
Random Forest           0.88       0.90         0.85      0.87
Gradient Boosting       0.87       0.89         0.84      0.86
SVM                     0.82       0.80         0.85      0.82

Best-Fit Model:
Model: Random Forest
Reasons: Highest F1-Score (0.87), Precision (0.90), and Recall (0.85)
```

Do It Yourself - 2

- Try to optmise the classification models based on breastcancer.csv using cross-validation method.

Cross-Validation Classification Model Results

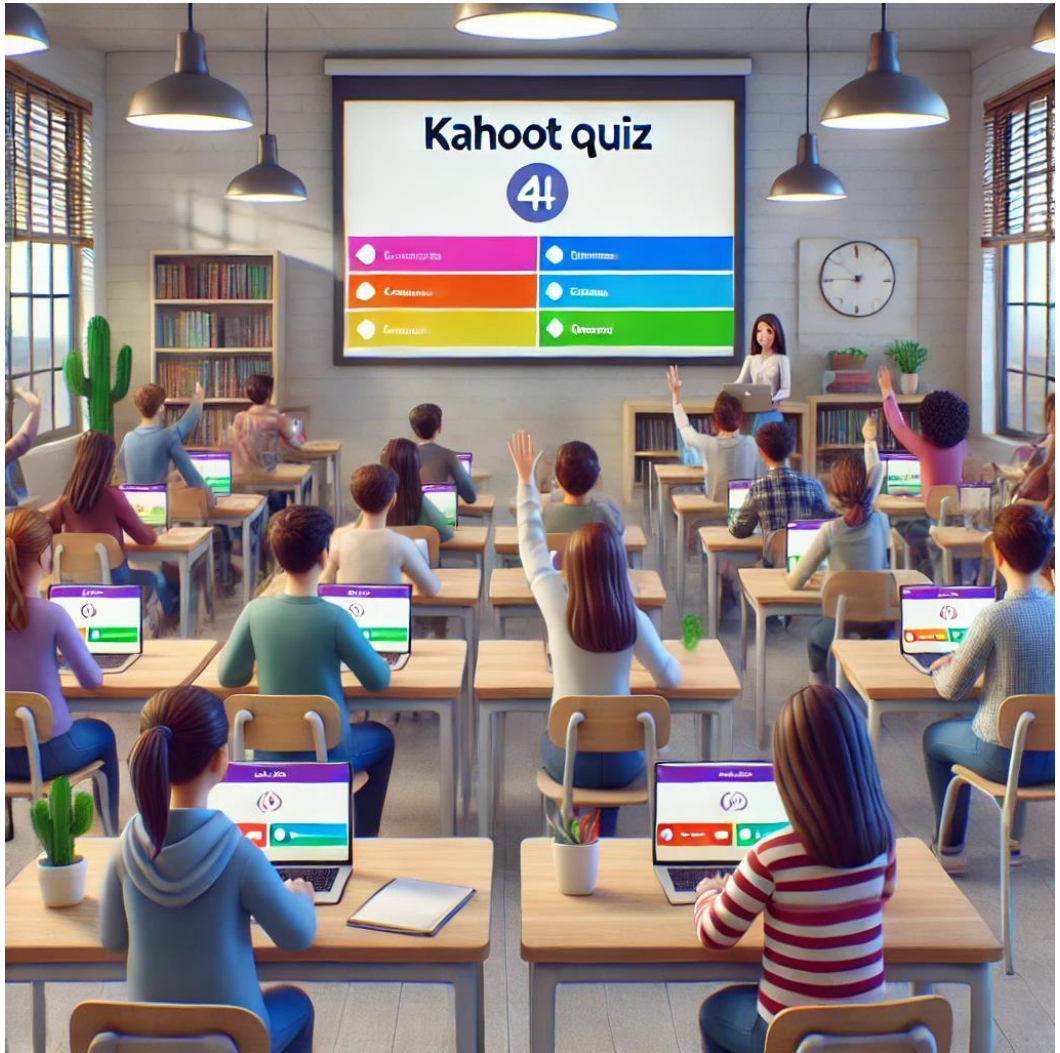
	Model	Confusion Matrix	Errors (FP + FN)	Accuracy	First 5 Predictions
1	Logistic Regression	[[434 10] [11 228]]	21	0.9692532942898975	[2 4 2 4 2]
2	K-Nearest Neighbors	[[437 7] [8 231]]	15	0.9780380673499268	[2 4 2 4 2]
3	Naive Bayes	[[424 20] [6 233]]	26	0.9619326500732065	[2 4 2 4 2]
4	Decision Tree	[[428 16] [19 220]]	35	0.9487554904831625	[2 4 2 4 2]

After optimization, KNN is the best-fit model.

Conclusion

- Logistic Regression is favored for its simplicity and interpretability in binary classification.
- KNN provides flexibility for non-linear boundaries but is computationally intensive for large datasets.
- Naïve Bayes is efficient and robust in high-dimensional spaces, especially for text classification.
- Decision Trees offer intuitive and visual decision-making but need techniques to prevent overfitting.
- Understanding each algorithm's strengths and limitations helps data scientists choose the most suitable method, ensuring accurate and efficient predictive models.

Kahoot Time!



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Thank You